

Mathematics A

Unit 2 • Chapter OL-T45 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

6 classified items • 28 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T45 • Expertise Q16+ • 6 items • 28 marks

Chapter	Items	Marks	Starts
01. OL-T45 Volume & Surface Area	6	28	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Volume & Surface Area

Unit 2 • Expertise • 28 marks • 6 item(s)

June 2024 | 4MA1-1H Q23 | 5 marks

Find volume or surface after removing a solid

November 2023 | 4MA1-1H Q21 | 5 marks

Find volume of a prism from its cross-section

June 2022 | 4MA1-1H Q22 | 5 marks

Find volume of a joined composite solid

January 2020 | 4MA1-2H Q26 | 6 marks

Find cone surface area or recover its sector

January 2021 | 4MA1-2H Q17 | 4 marks

Find surface area of a cuboid or simple prism

January 2023 | 2H Q21 | 3 marks

Find sphere or hemisphere surface area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q23

5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

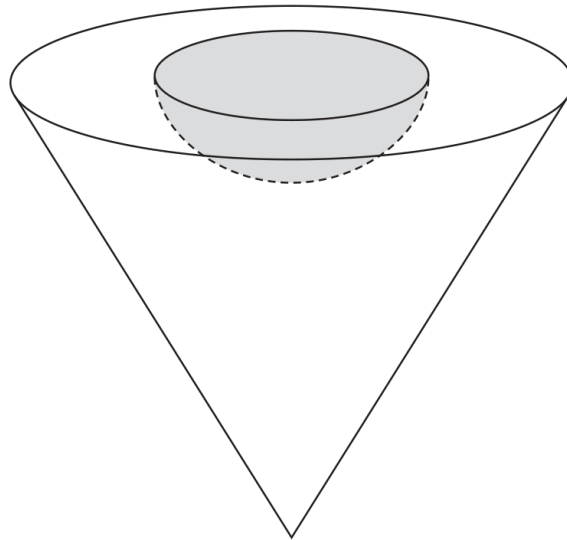


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q23

5 marks

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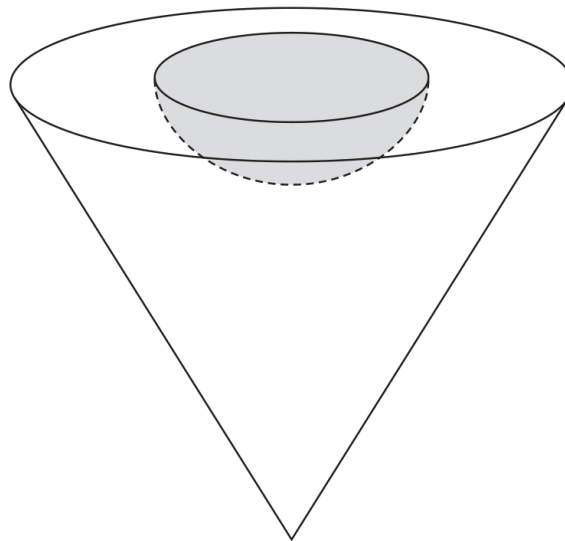


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..... cm²

Cone with a hemisphere removed**5 marks****Find the scale**

$$50\pi x^3 - \frac{16}{3}\pi x^3 = 6948\pi \Rightarrow x^3 = 10422/67.$$

Removed surface

$$3\pi(2x)^2 = 12\pi x^2 = 12\pi(10422/67)^{2/3} = 1090.390\dots$$

Final answer

$$1090 \text{ cm}^2$$

4MA1-1H Q21 demand

5 marks

21 Here is a triangular prism $ABCDEF$

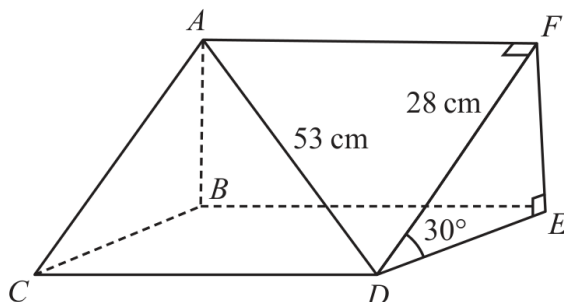


Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

WORKING SPACE

4MA1-1H Q21 demand

5 marks

21 Here is a triangular prism $ABCDEF$

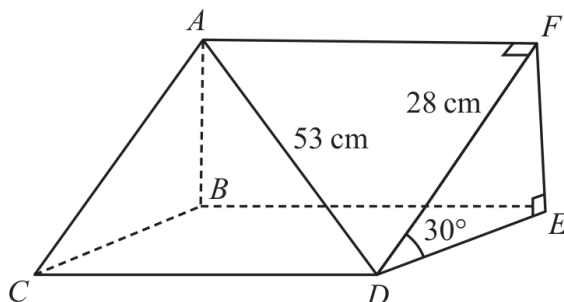


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..... cm^3

Volume of a triangular prism

5 marks

Demand – Volume of a triangular prism

Resolve the triangular end

$$FE = 28 \sin 30^\circ = 14, \quad DE = 28 \cos 30^\circ = 14\sqrt{3}.$$

Find prism length

$$AF = \sqrt{53^2 - 28^2} = 45.$$

Calculate volume

$$V = \frac{1}{2}(14)(14\sqrt{3})(45) = 4410\sqrt{3} \approx 7638.34.$$

Final answer7638 cm³

4MA1-1H Q22 Q22

5 marks

22 A solid is made from a cone and a hemisphere.

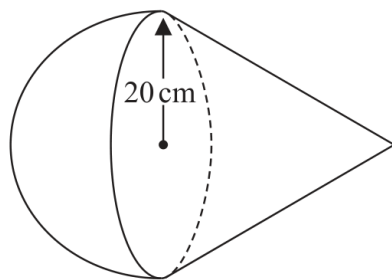


Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi\text{cm}^2$

The volume of the solid is $k\pi\text{cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q22 Q22

5 marks

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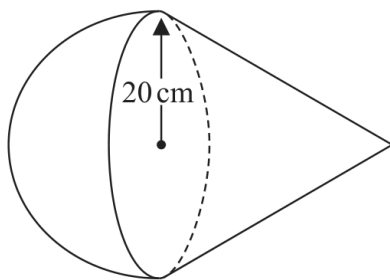


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$k = \dots\dots\dots$

Chapter: Volume & Surface Area

5 marks

Q22 – Chapter: Volume & Surface Area

Family: Resolve composite solids and volume-unit constraints • Variant: Find volume of a joined composite solid

Method

For the cone, curved surface area $\pi r l = 580\pi$ gives $\pi(20)l = 580\pi$.

Why: Use the given radius in the curved-surface-area formula.

Method

The cone slant height is $l = 29$ cm.

Why: Cancel π and divide by 20.

Method

The cone height is $\sqrt{29^2 - 20^2} = \sqrt{441} = 21$ cm.

Why: The radius, height, and slant height form a right triangle.

Method

The solid volume is $(\frac{2}{3})\pi(20)^3 + (\frac{1}{3})\pi(20)^2(21)$.

Why: Add the hemisphere volume to the cone volume.

Method

This is $(\frac{16000}{3} + \frac{8400}{3})\pi = (\frac{24400}{3})\pi$, so $k = 24400/3$.

Why: Combine the exact coefficients of π .

Final answer

k: $k = 24400/3$

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

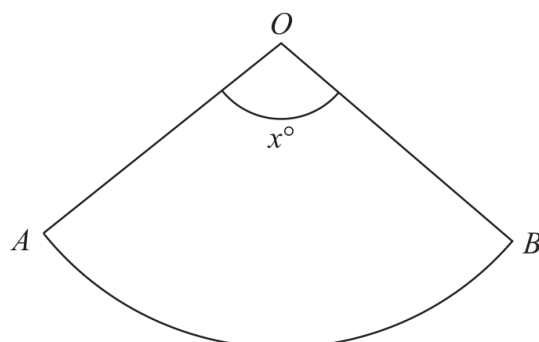


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

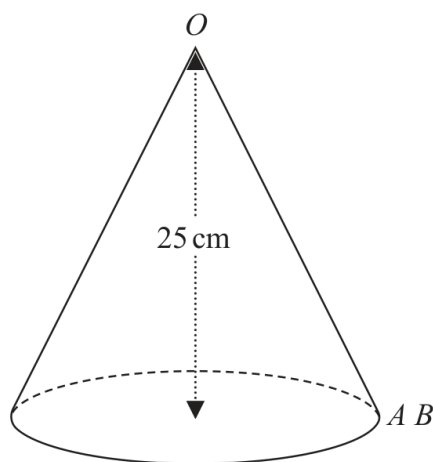


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm .

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

Worked solution

January 2020 | Question 26 | 6 marks

Family: Use sphere and cone formulae • Variant: Find cone surface area or recover its sector

Method / working

1. Cone volume gives $r^2 = 4800 / (25\pi)$, so $r \sim 7.8176$ cm.
2. Slant height $l = \sqrt{r^2 + 25^2} \sim 26.1938$ cm.
3. Sector arc equals base circumference: $(x/360)2\pi l = 2\pi r$.
4. Cancel 2π and rearrange $x = 360r/l$.
5. $x \sim 107.443^\circ$.
6. Nearest whole degree: $x = 107^\circ$.

Final answer: 107°

17 The diagram shows a solid prism $ABCDEFGH$.

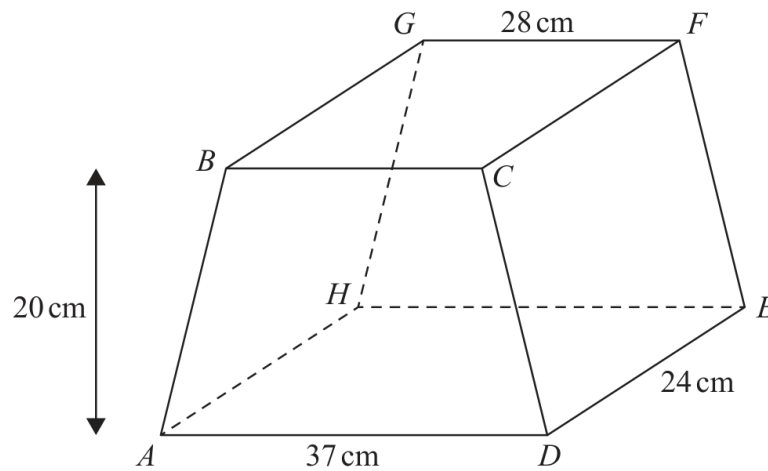


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37 \text{ cm} \quad GF = 28 \text{ cm} \quad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

(a) Work out the total surface area of the prism.

..... cm^2
(4)

Worked solution

January 2021 | Question 17 | 4 marks

Family: Find surface area of prisms and cylinders • Variant: Find surface area of a cuboid or simple prism

Method / working

1. Area of trapezium ABCD = $\frac{1}{2}(37+28) \times 20 = 650 \text{ cm}^2$.

2. Sloping side AB = $\sqrt{20^2 + 4.5^2} = 20.5 \text{ cm}$.

3. Add two trapezia and four rectangular faces: $2 \times 650 + 2 \times 20.5 \times 24 + 37 \times 24 + 28 \times 24$.

4. Total surface area = 3844 cm^2 .

Final answer: 3844 cm^2

21 Given that the surface area of a sphere is $49\pi\text{cm}^2$

find the volume of the sphere.

Give your answer correct to the nearest integer.

..... cm^3

(Total for Question 21 is 3 marks)

Answer reference | January 2023 | Question 21

Family: Use sphere and cone formulae • Variant: Find sphere or hemisphere surface area

Question	Working	Answer	Mark	Notes
21	$r = \sqrt{\frac{49\pi}{4\pi}}$ oe (= 3.5)		3	M1
	[volume =] $\frac{4}{3} \times \pi \times 3.5^3$			M1
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	180		A1 awrt 180
				Total 3 marks

Worked solution

January 2023 | Question 21 | 3 marks

Family: Use sphere and cone formulae • Variant: Find sphere or hemisphere surface area

Official final mark-scheme pages are embedded immediately after this question.